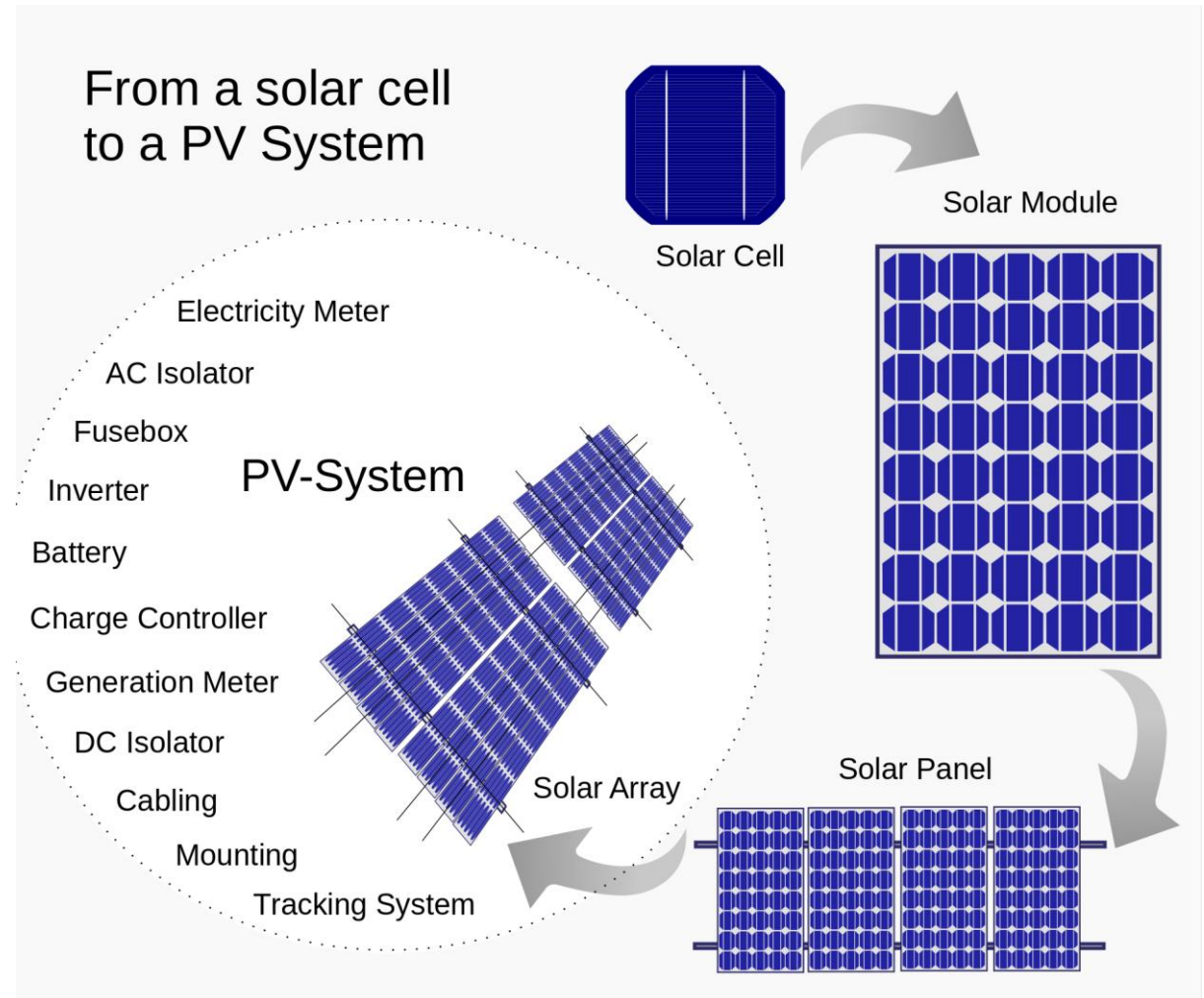


Fundamentals of Photovoltaics (EE338)

Lecture 1: Introduction

What is photovoltaics (PV)?

- PV is a term which covers the conversion of light into electricity using semiconducting materials that exhibit the photovoltaic effect.
- A typical PV system employs solar panels, each comprising a number of solar cells, which generate electrical power.
- Other components of a PV system; DC-AC inverter, battery, mounting, concentrator, tracking system, etc.,

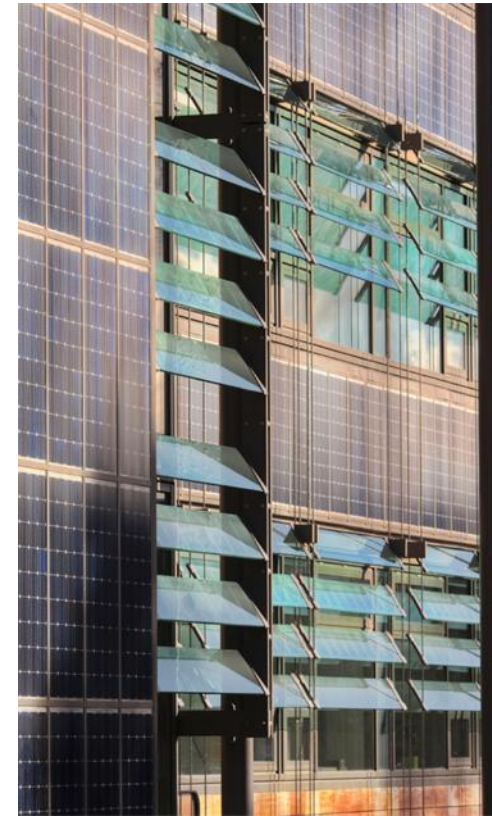


PV installation type: **ground-mounted**, **rooftop mounted**, **wall mounted**

The mount can be fixed, or use of **solar tracker** to follow the sun across the sky but 10-25% is lost if the tracking system is not used.

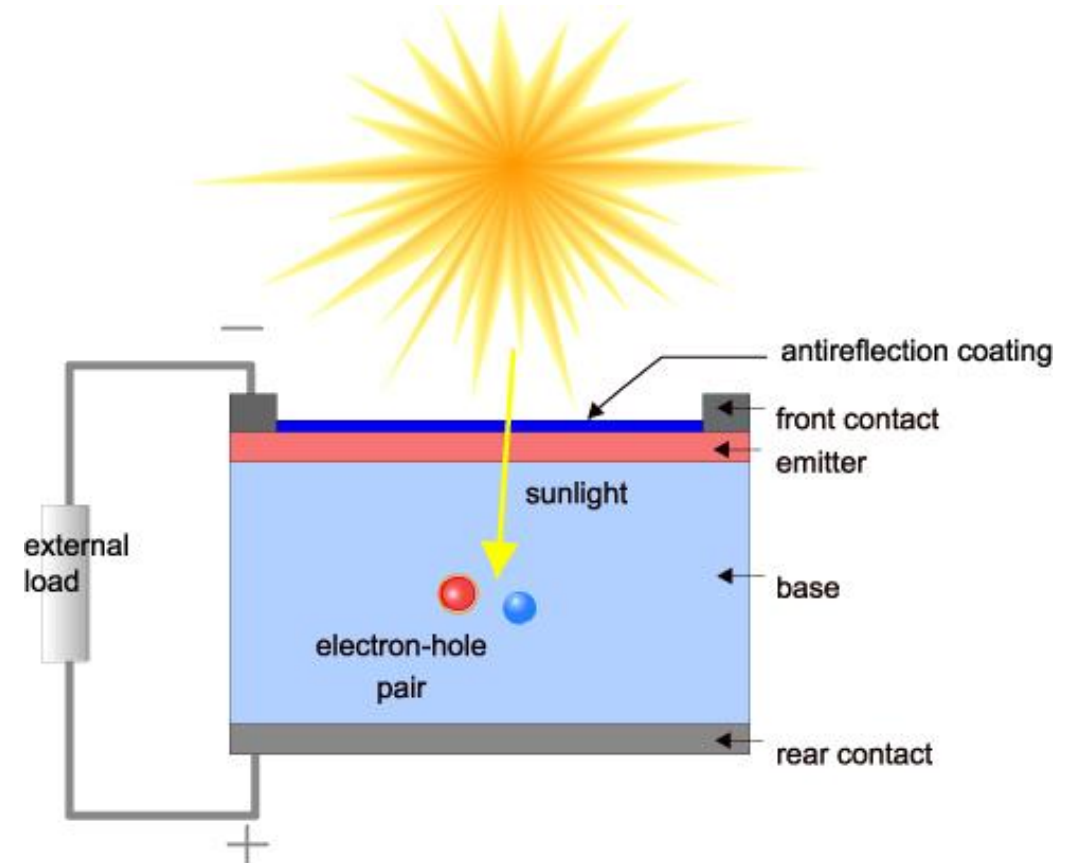
The function of a tracker is to keep the plane of solar array perpendicular to the sun in both azimuth and elevation. (disadvantage – cost)

Standalone or Grid-connected



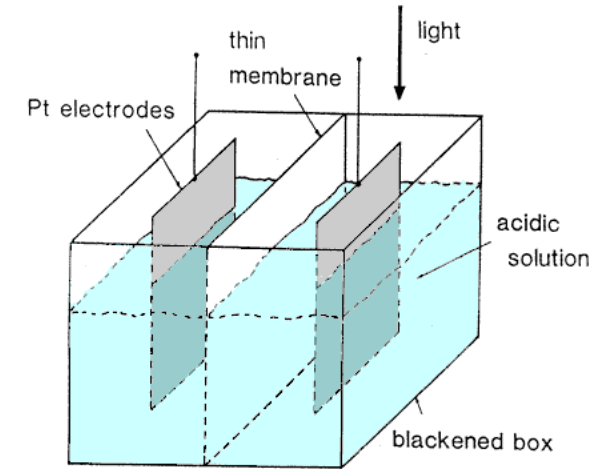
How does a solar cell work?

- The light with photon energy higher than the bandgap of a semiconductor is absorbed by the semiconductor
- Electron-hole pairs or excitons are generated upon light absorption
- The electron-hole pairs or excitons are separated by internal electric field or band offset and become free charge carriers
- The free charge carriers are collected at the electrodes.



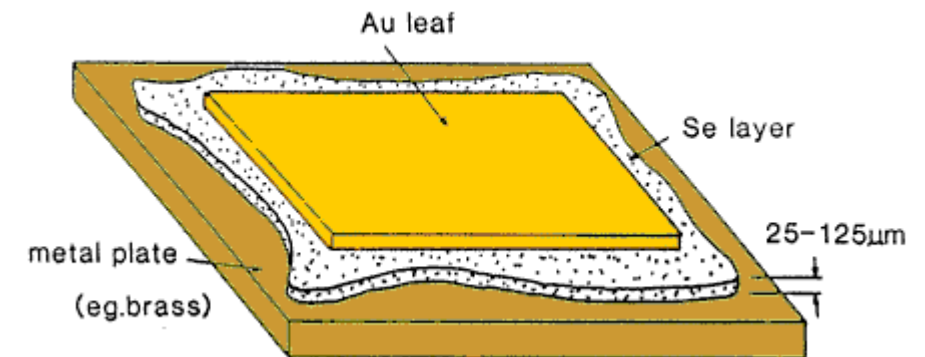
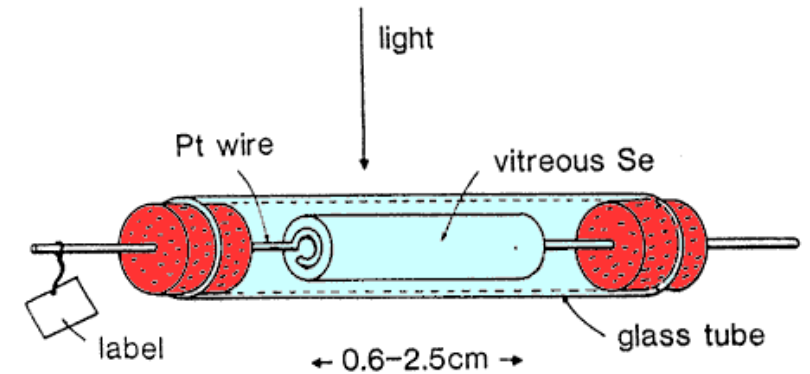
History of Solar Cell

- 1839, Edmond Becquerel (French Physicist); discovered at age 19
- silver chloride was placed in an acidic solution and illuminated while connected to platinum electrodes, generating voltage and current.
- Liquid-state photoelectrochemical cell



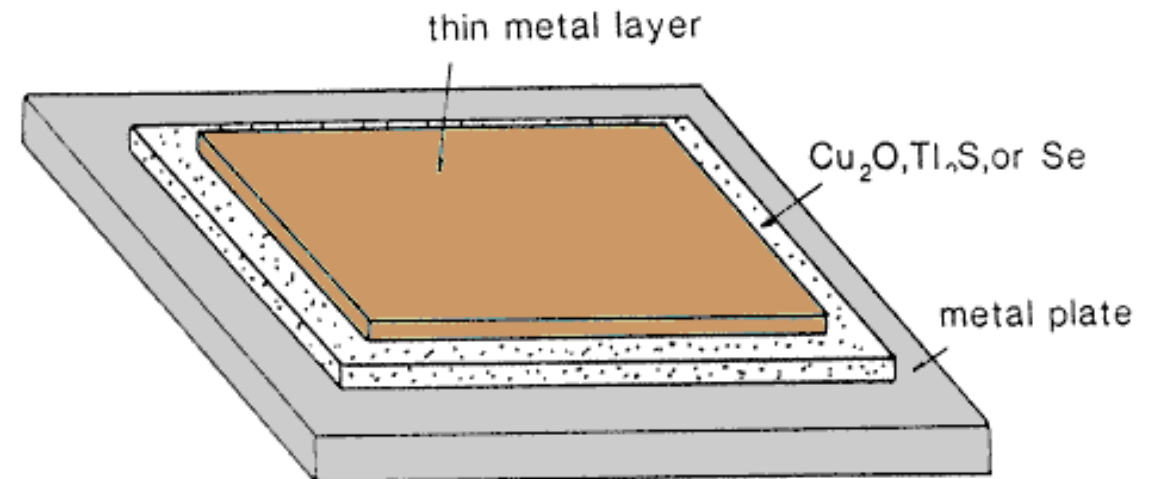
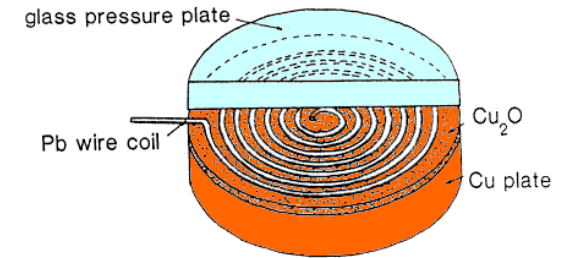
History of Solar Cells

- 1870s, PV effect in Selenium by W. Smith, W. G. Adam and R. E. Day
 - First demonstration of PV effect in solid-state system
- 1883, C. E. Fritts
 - Selenium was sandwiched between two different metal plates
 - His observation was skeptical because there was no quantum theory then
 - Werner Siemens verified his observation
 - The efficiency is less than 1%



History of Solar Cells

- 1930s
 - Grondahl, copper-cuprous oxide photovoltaic cells
 - Bergmann, selenium cell
 - Nix, thalious-sulphide cell
- 1954, Bell Lab invented p-n junction silicon single crystal solar cell with 6% efficiency
- 1958, the first application of solar cell in satellite (Vanguard I)



Motivation for PV Technology

- **Energy Security Issue**

- Rapid growth in energy demand
- Falling oil production rate globally (controversial)

- **Environmental Issue**

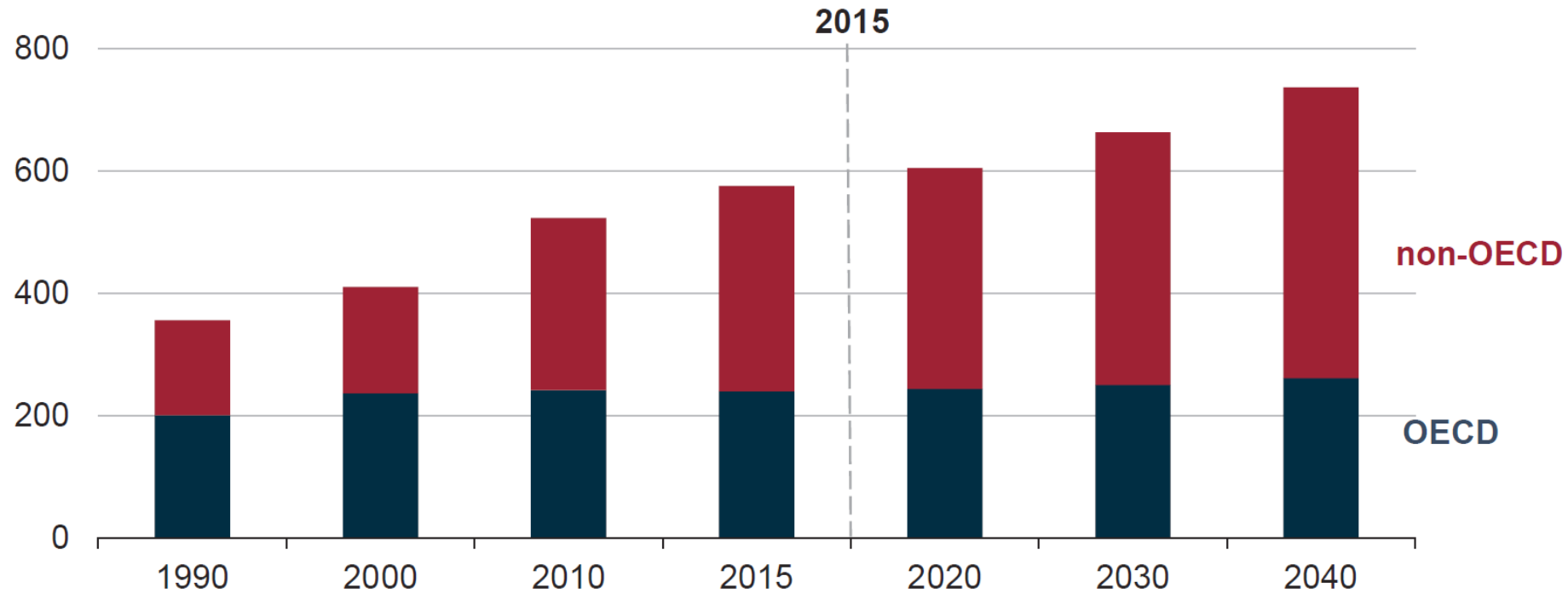
- Increasing greenhouse gas level (CO_2 concentration 280 ppm in 18th century, current level 380 ppm)
- Global warming and sea level rise

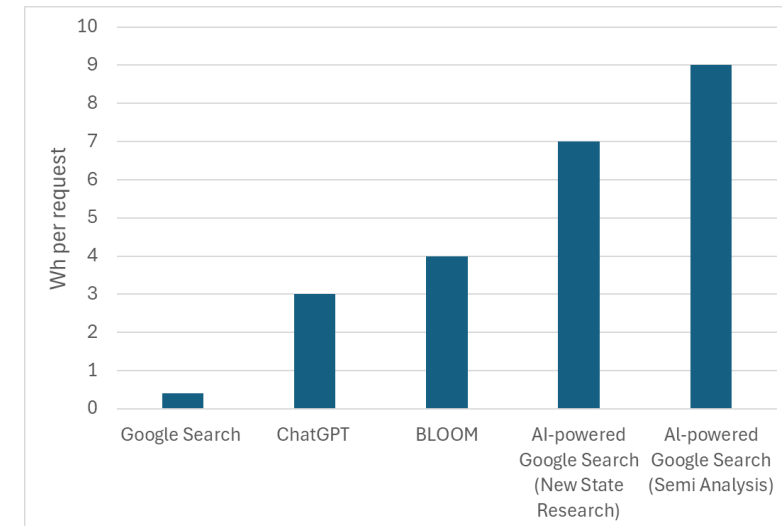
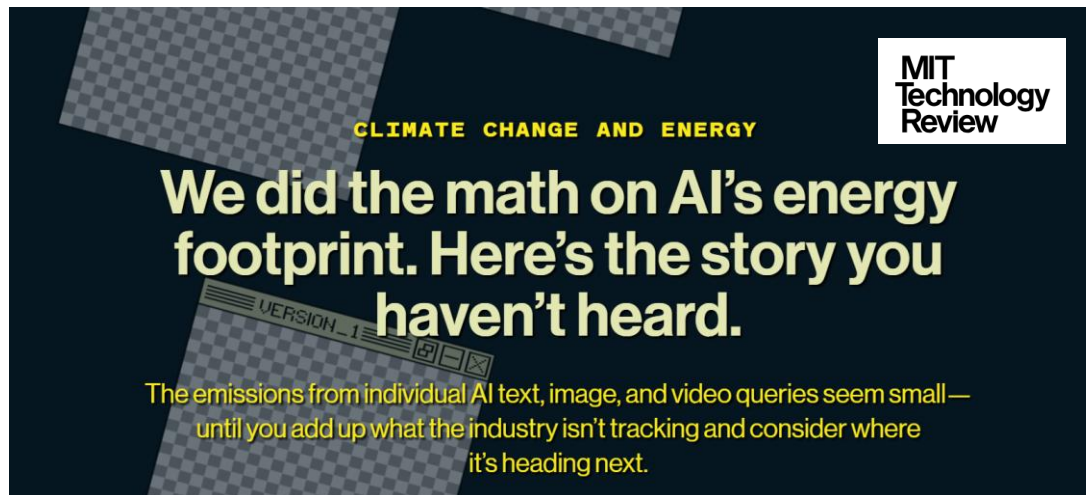
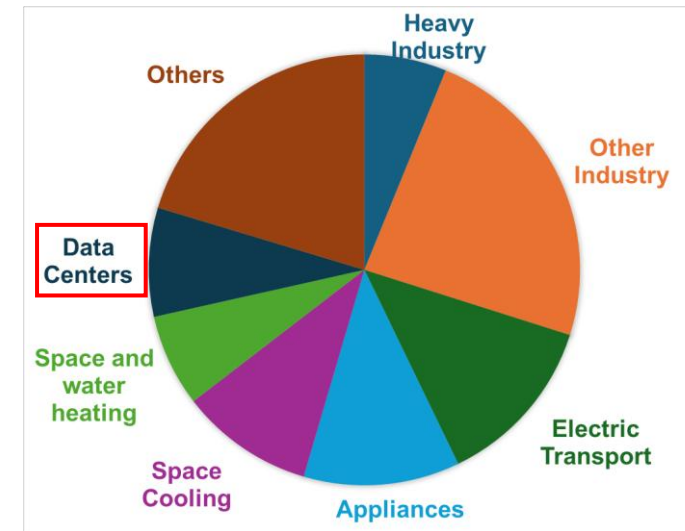
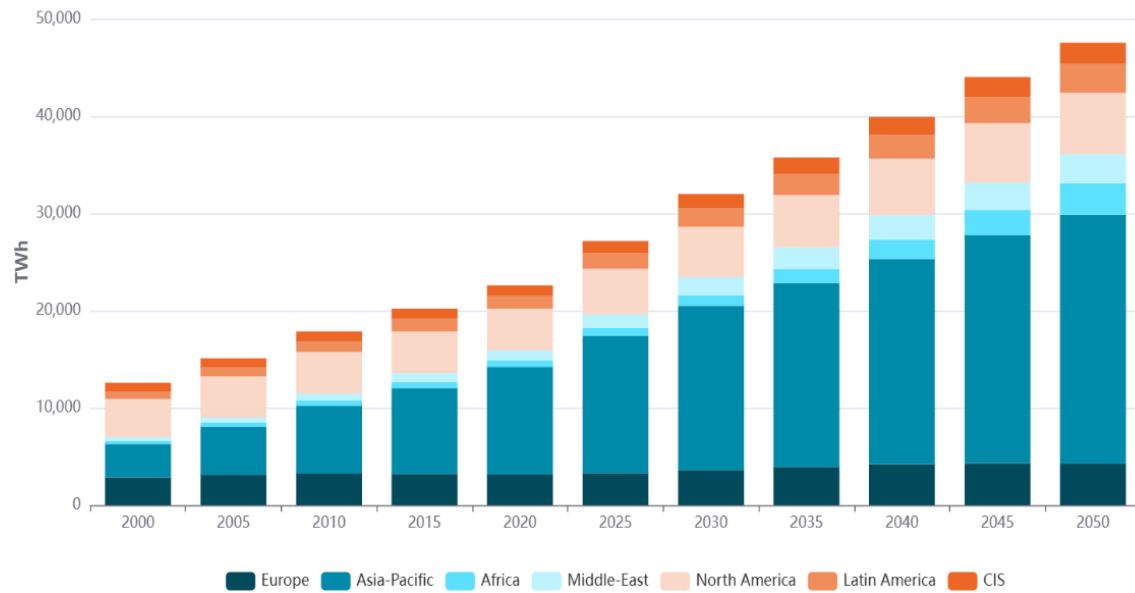


Motivation for PV Technology

World energy consumption rises 28% between 2015 and 2040

World energy consumption
quadrillion Btu

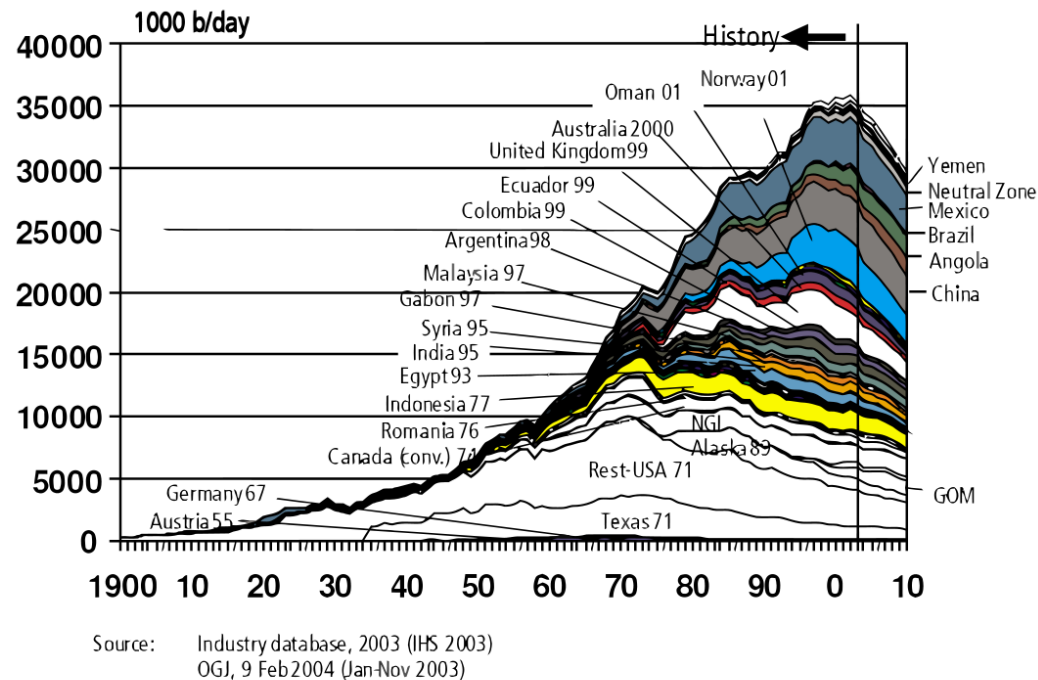




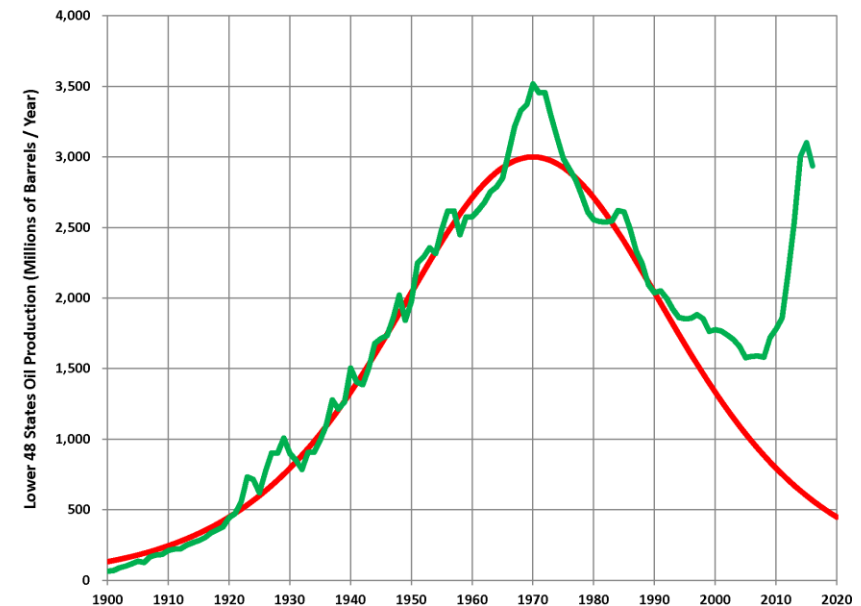
❑ Electricity consumption almost doubles worldwide and reaches ~ 48,000 TWh in 2050, representing an average growth of 2.4% annually.

Hubbert peak theory and falling oil production rate

- The **Hubbert peak theory** says that for any given geographical area, from an individual oil-producing region to the planet as a whole, the rate of petroleum production tends to follow a **bell-shaped curve**.

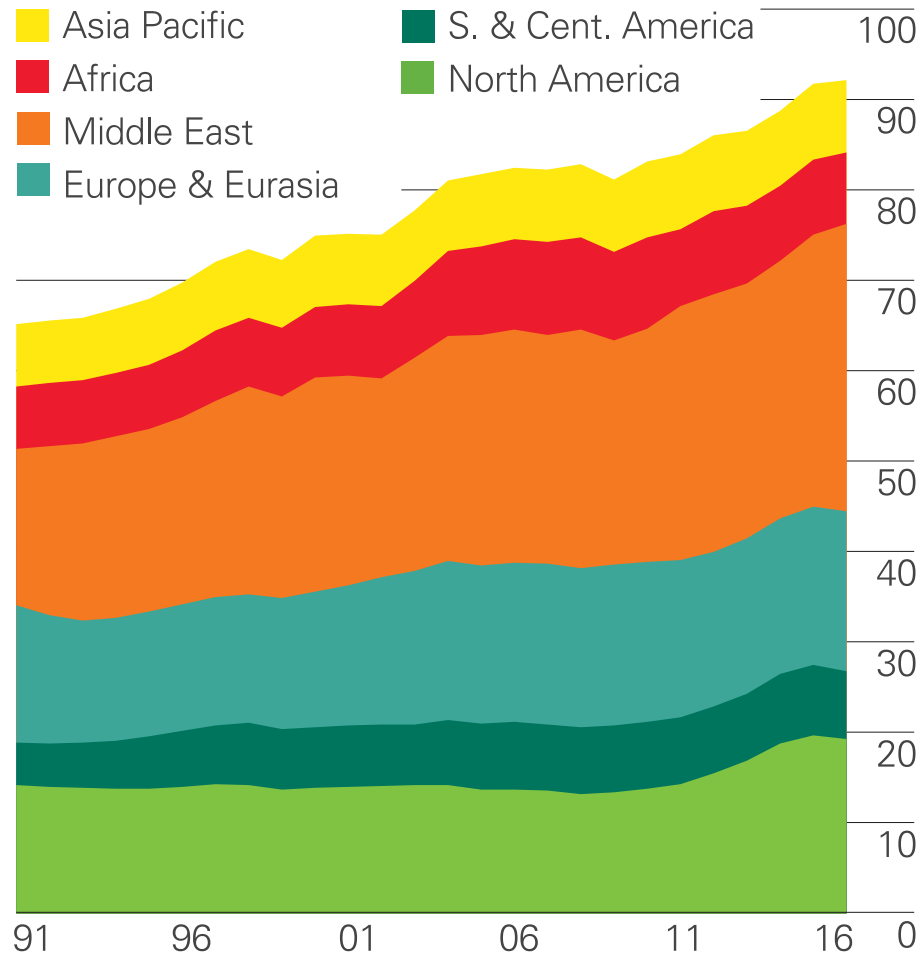


2004 U.S. government predictions for oil production other than in OPEC and the former Soviet Union

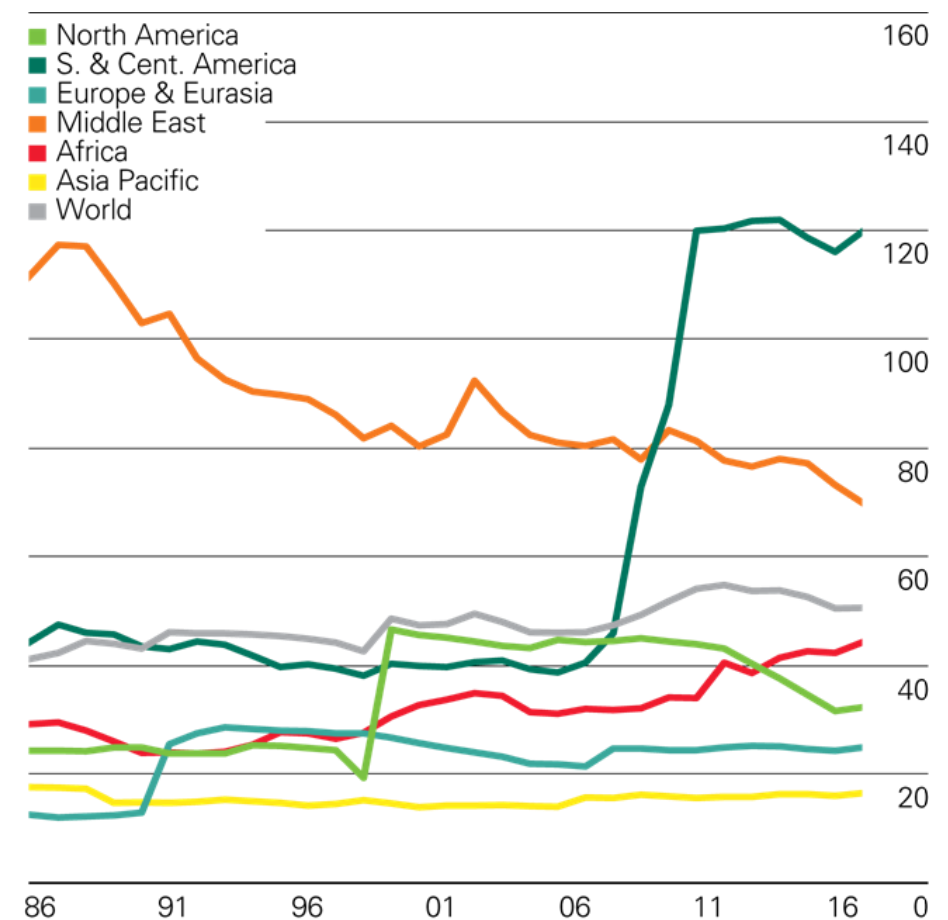


Hubbert's upper-bound prediction for US crude oil production (1956), and actual lower-48 states production through 2014

Oil production by region

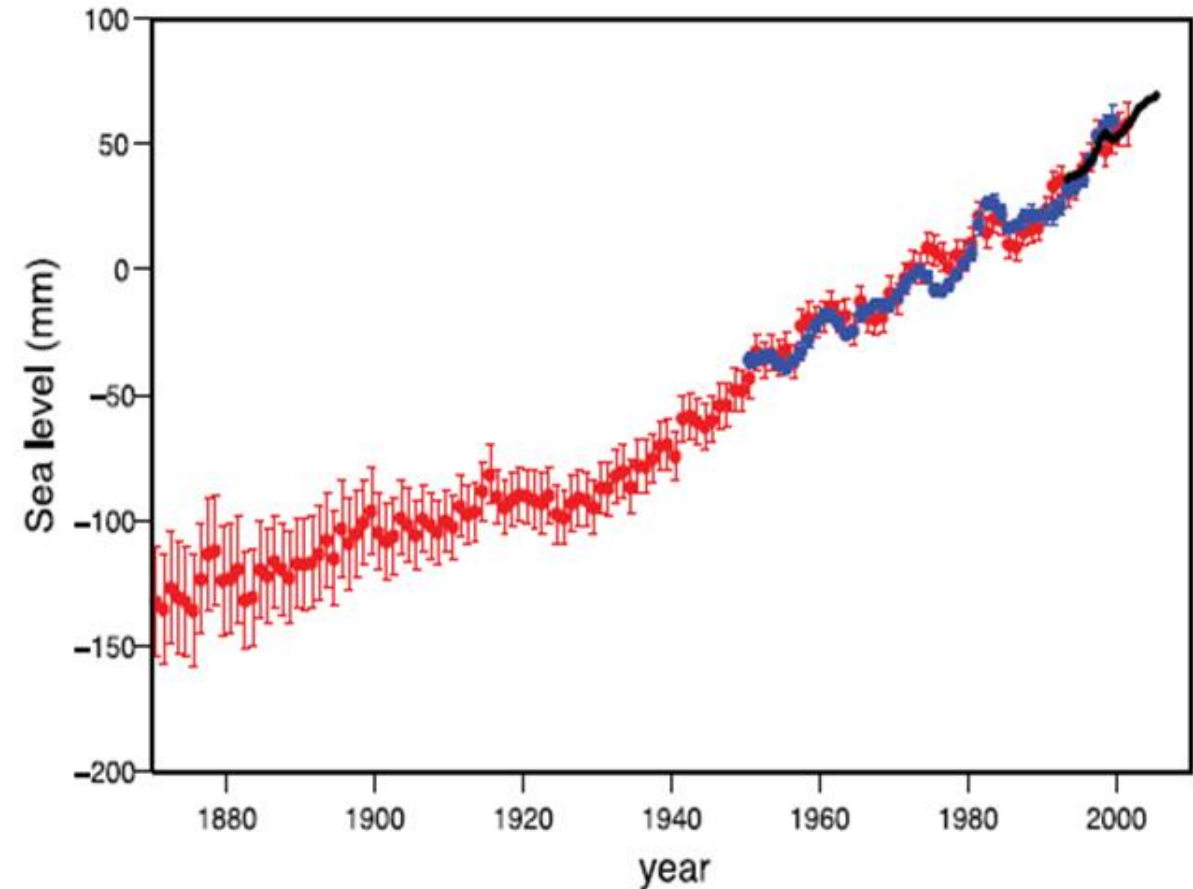
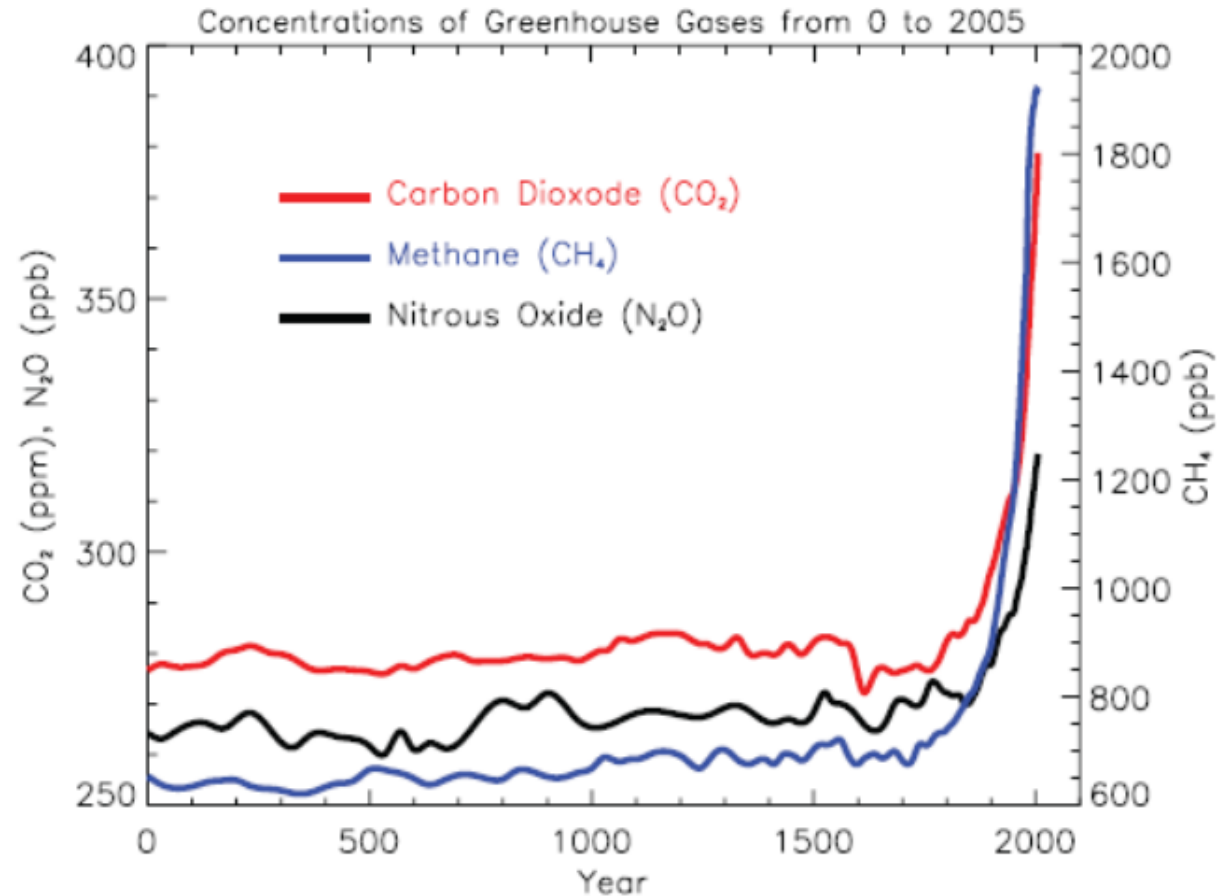


Oil reserves-to-production (R/P) ratios



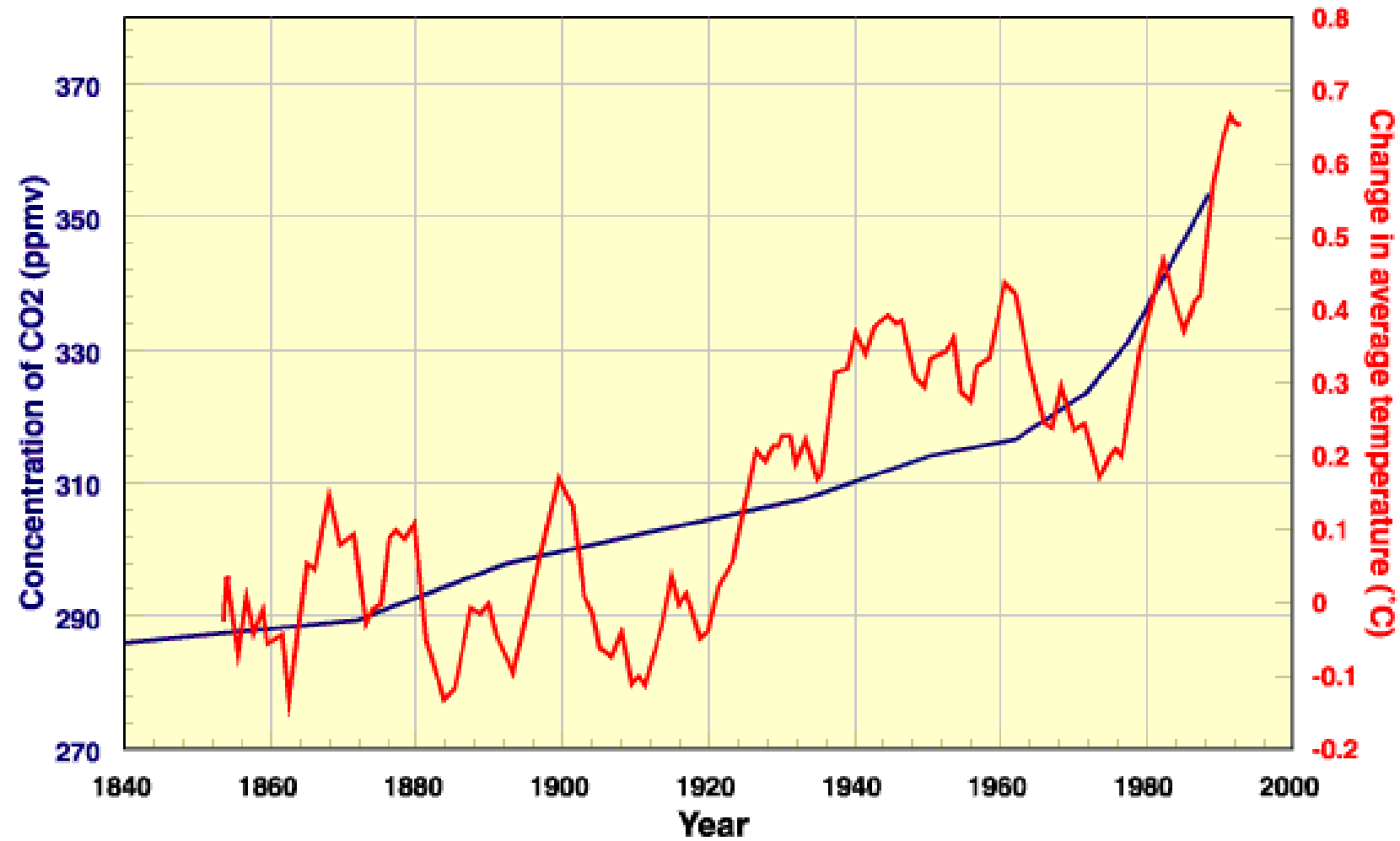
Source: www.bp.com

Motivation for PV Technology



Source: Intergovernmental Panel on Climate Change (IPCC)

Motivation for PV Technology



Solar Energy is **UNIQUE** because

- Abundance
- Environmentally Friendly (No CO₂ and greenhouse gas emission once installed)
- Geographically diverse
- No impact on biodiversity and wildlife

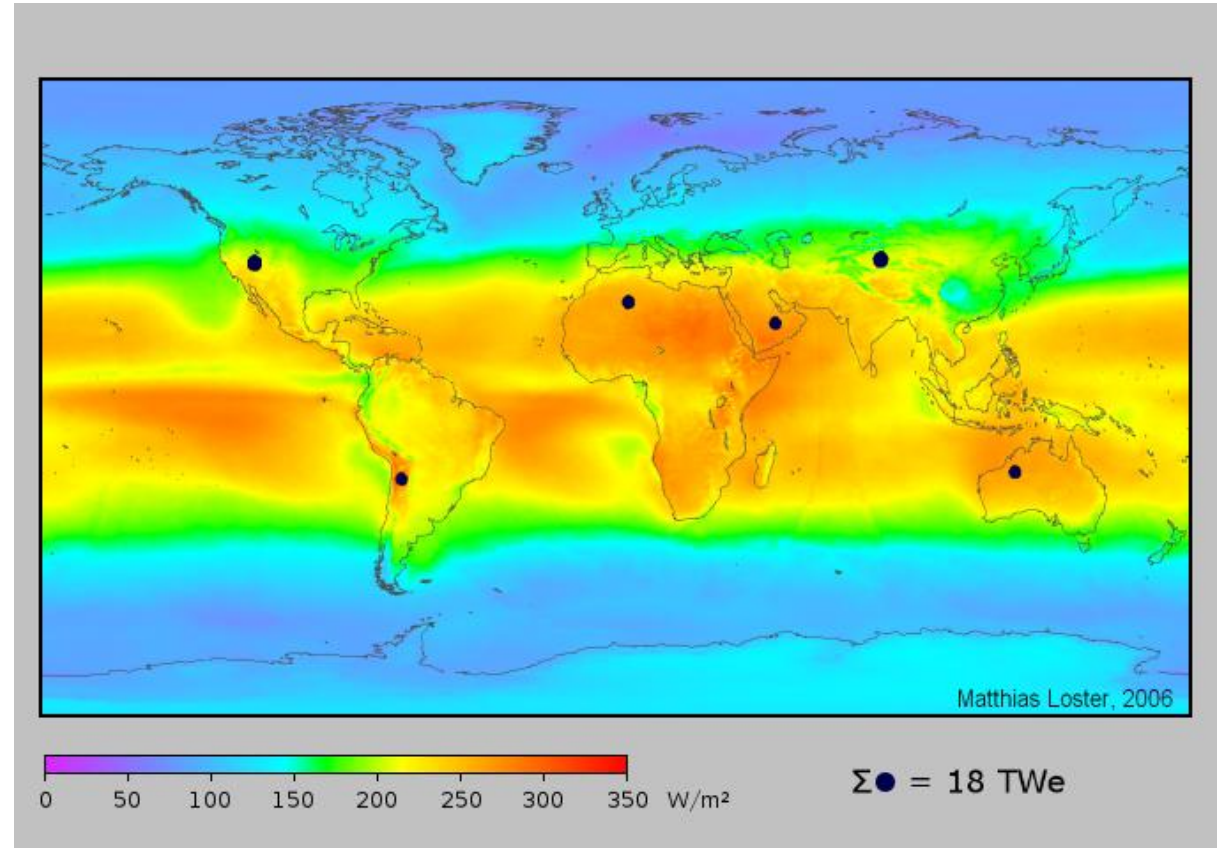


Abundance

- At the top surface of the earth's atmosphere that is perpendicular to the direction of the sun, the average irradiance or solar intensity is **1353 W/m²**.
- With this rate, the energy from sunlight impinges on the Earth in **one hour** is more than all of the energy consumed by humans in **an entire year**
- At sea level, on a day with clear sky at a terrestrial position, the solar energy received is at the level of 700-900 W/m².
- The annual average horizontal surface irradiance, however, is approximately **170 W/ m²** due to intermittency and changes during the day and with seasons and also losses in passing through the earth's atmosphere.
- Nonetheless, when 170 W/m² is integrated over 1 year, the resulting **5.4GJ** that is incident on just 1 m² is equivalent to the energy extracted from **one barrel of oil** or **140 m³ of natural gas**.

Geographically diverse

- Most of the regions of the world fall within the favorable regions and receive above the average irradiance of 170 W/m^2
- Red Sea area, $\sim 300 \text{ W/m}^2$ (the highest irradiance)
- Australia, $\sim 200 \text{ W/m}^2$
- USA, $\sim 185 \text{ W/m}^2$
- United Kingdom, $\sim 105 \text{ W/m}^2$



small black dots are the area required to replace the world primary energy supply with solar electricity, 18 TW

Solar Irradiation in China

- Region I (e.g. Tibet)
Higher than southwestern desert region of United States
- Region II (e.g. Beijing)
similar to Madrid, Spain although it is not as high as southwestern United States
- Region III (e.g. Shanghai)
lower than most of United States but well above Germany

TABLE 8.1. Solar Irradiation Resources in China on a Horizontal Surface

Level of Irradiation	Region Number	Annual Irradiation Max (MJ/m ²)	Annual Irradiation Min (kWh/m ²)	Daily Irradiation Average (kWh/m ²)
Very high	I	≥6300	≥1750	≥4.8
High	II	5040–6300	1400–1750	3.8–4.8
Average	III	3780–5040	1050–1400	2.9–3.8
Low	IV	<3780	<1050	<2.9



Solar Availability in China Relative to United States and Europe

Region	City and Latitude	Annual Average Hours per Day on Horizontal Surface	Annual Average Hours per Day at Fixed Tilt	Tilt Angle	Annual Average Hours per Day 1-Axis Tracking
China	Beijing (39.6)	4.3	4.8	42	
	Shanghai (31)	3.6	4.0	35	
	Xizang (Tibet)	6.0	6.7	30	
United States	Medford, Oregon	4.4	4.9	42	6.5
	Sacramento California	4.9	5.5	39	7.4
	Boston, Massachusetts	3.9	4.6	42	5.7
	Phoenix Arizona	5.7	6.5	33	8.6
Europe	Madrid (40), Spain		4.9	30	
	Lisbon (38), Portugal		5.4	30	
	Berlin (52), Germany		2.8	30	

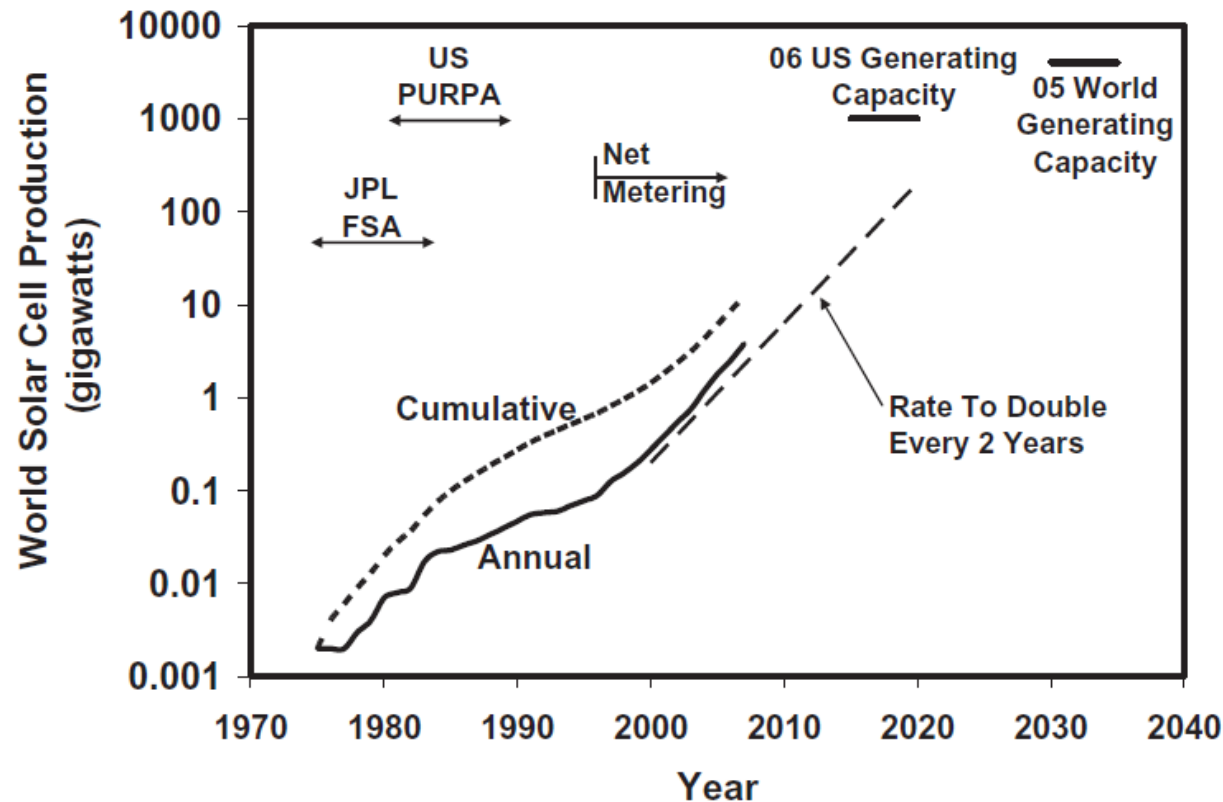
Current use and current potentials of selected renewable energy sources

Resource	Current use	Technical potential	Theoretical potential
Hydropower	9	50	147
Biomass energy	50	>276	2900
Solar energy	0.1	>1575	3 900 000
Wind energy	0.12	640	6000

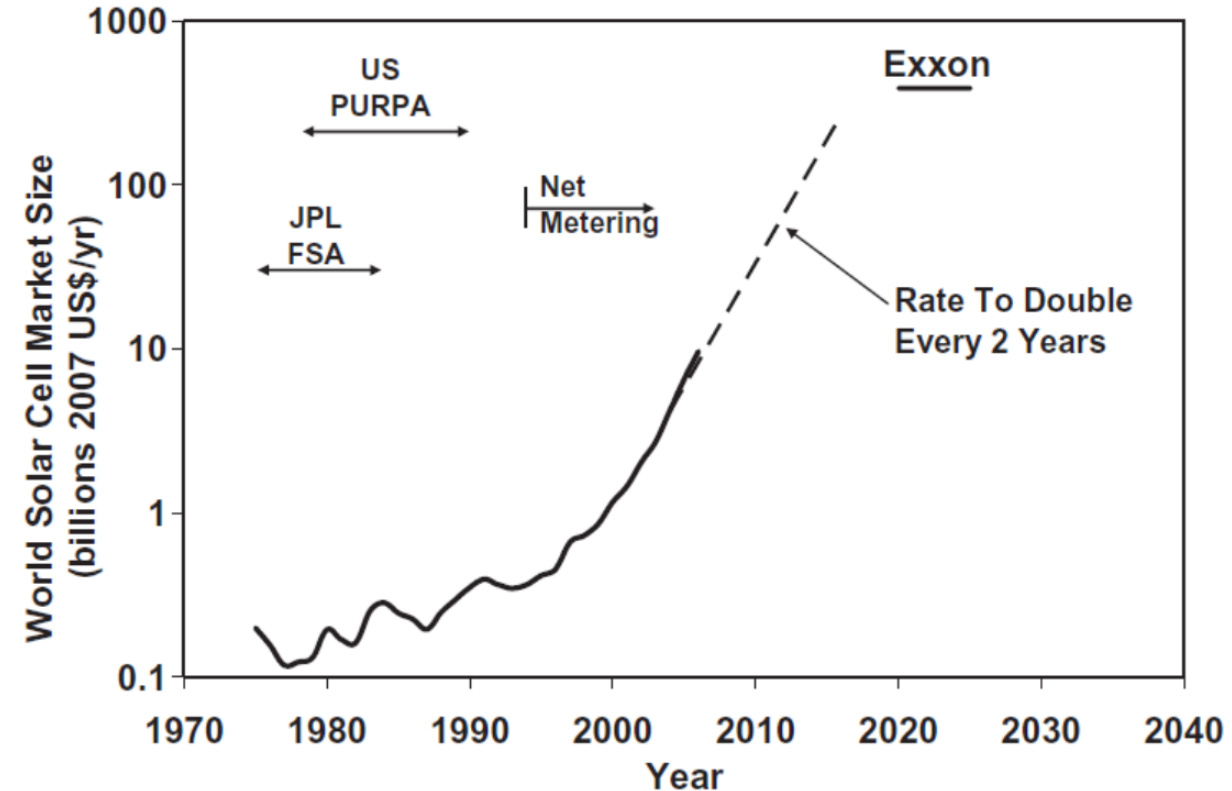
Units: exajoule per year

1 Exajoule = 10^{18} J

Market History and Public Policy

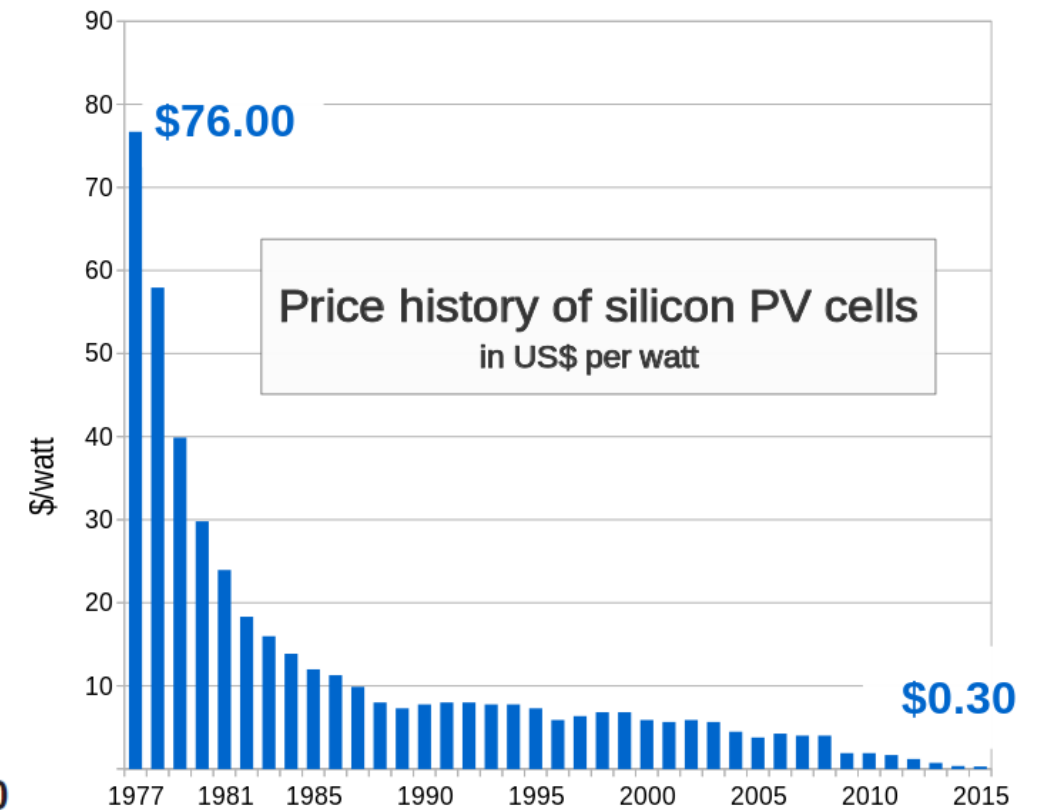
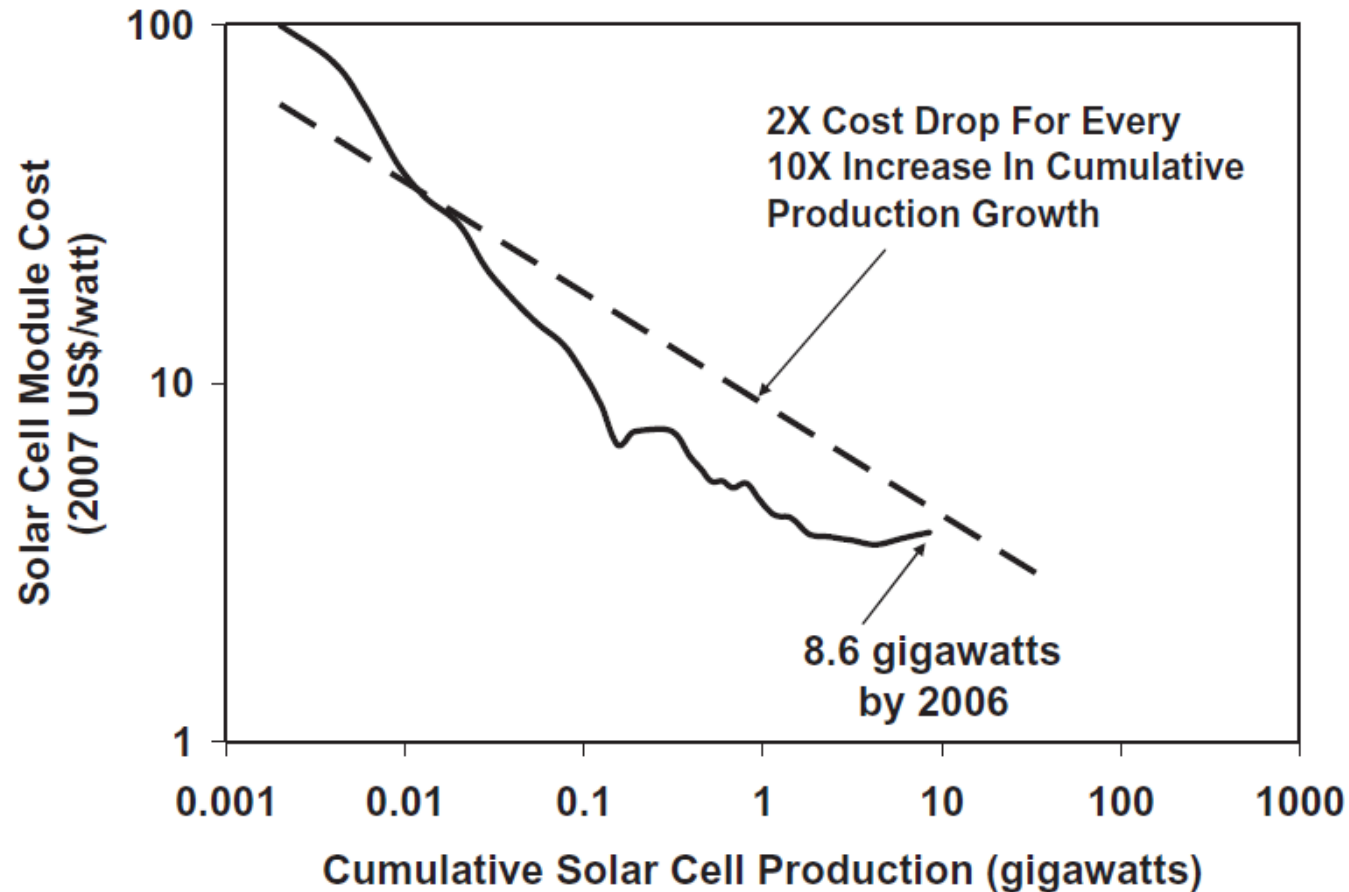


The world solar cell production levels, on both an annual and cumulative basis, since 1975, with comparisons to the U.S. and world total electricity generating capacities in 2006 and 2005, respectively, and with support and public policy program time intervals or the JPL FSA, the U.S. PURPA, and the net metering programs.



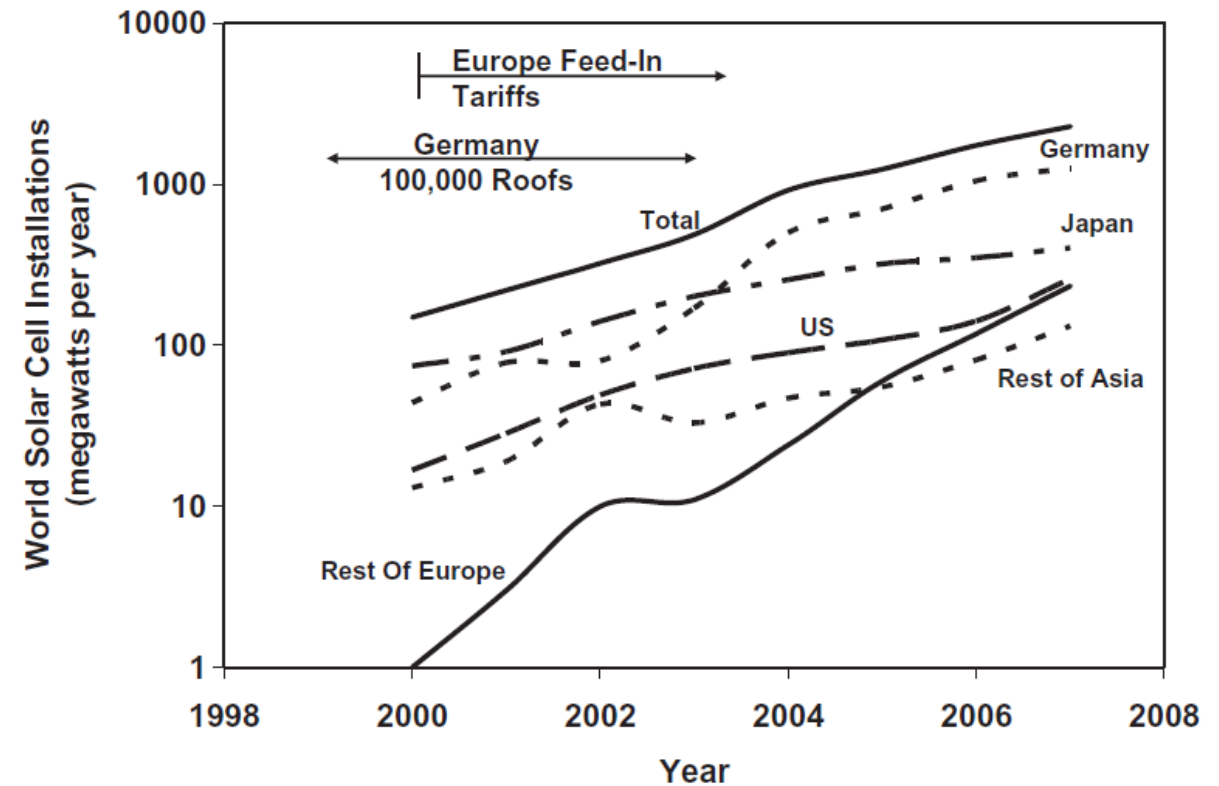
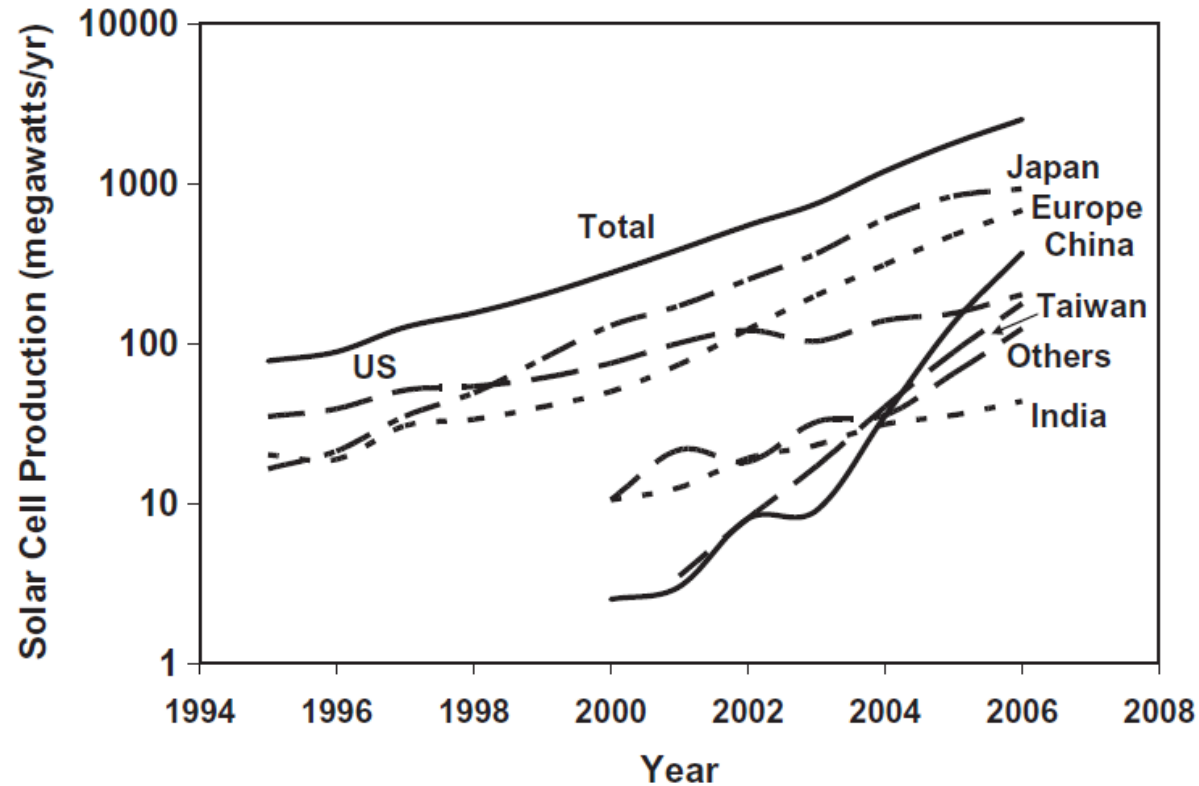
The world annual solar cell market size since 1975 with a comparison to the annual sales of the world's largest oil company, Exxon Mobile, in 2007.

Market History and Public Policy

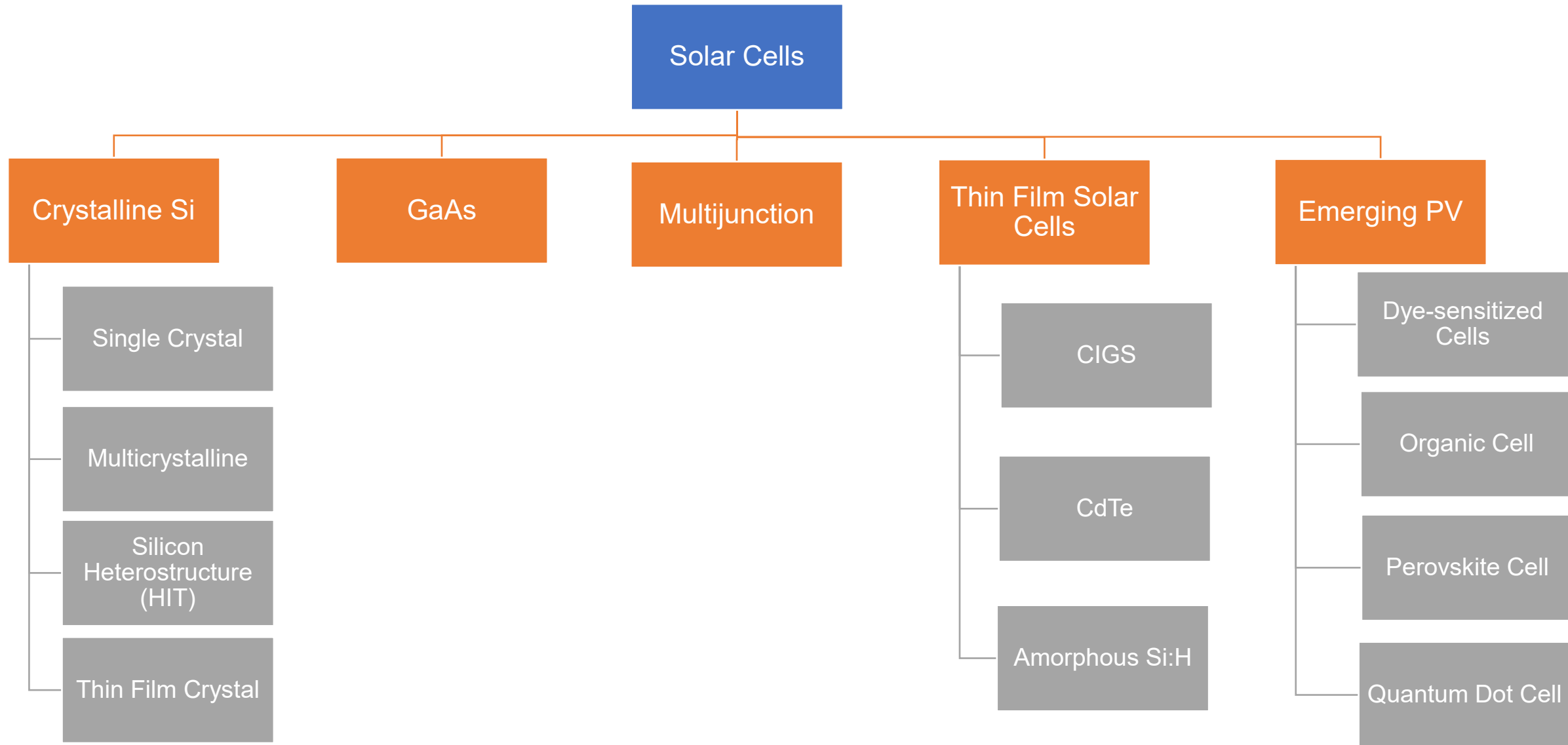


Source: Bloomberg New Energy Finance & pv.energytrend.com

Market History and Public Policy



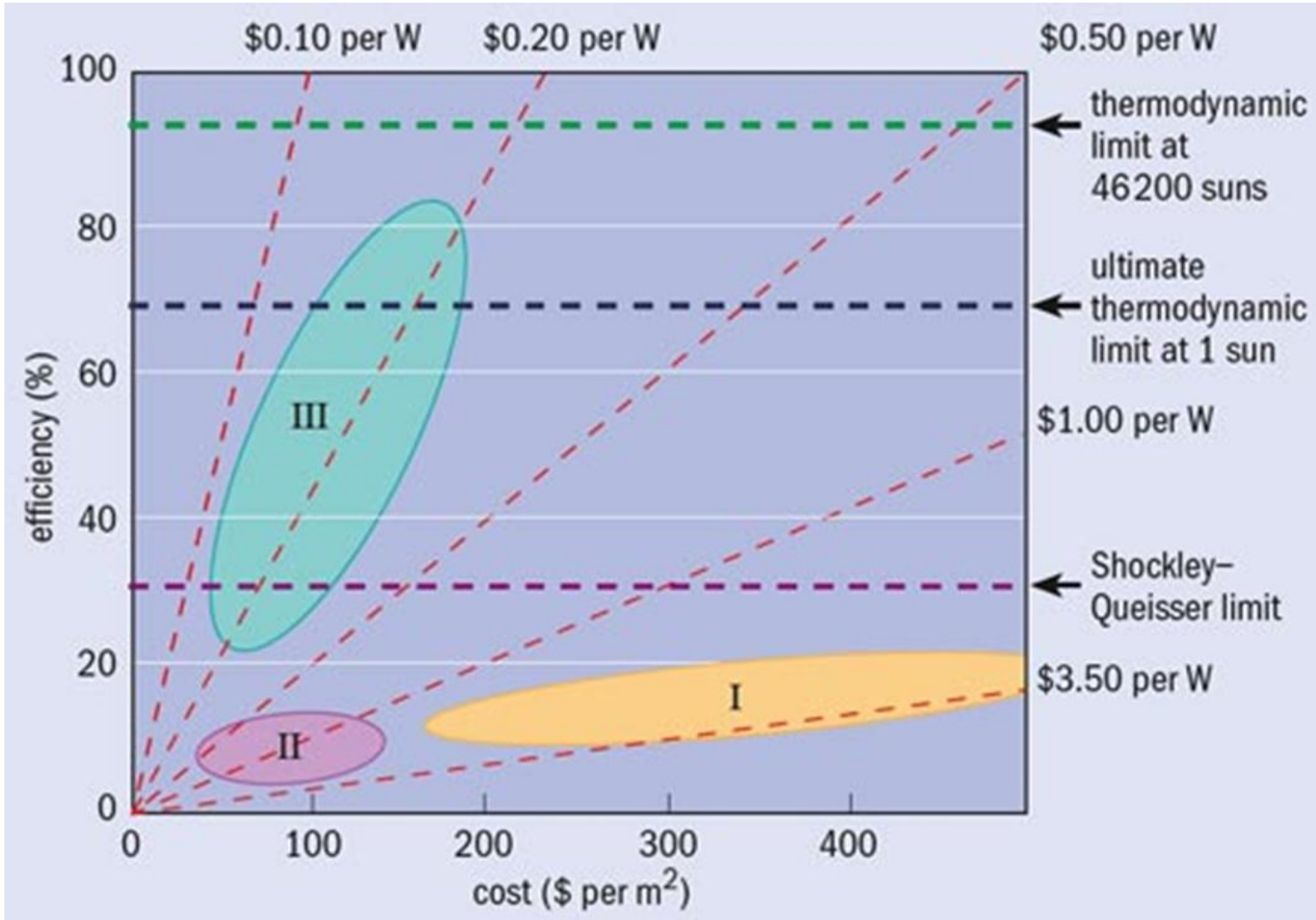
Classification of Solar Cells



NREL
NATIONAL RENEWABLE ENERGY LABORATORY



Efficiency Vs Cost



I – silicon p-n junction solar cells

II – thin film solar cells

III – Frequency Conversion Technology, Hot-carrier effect, other multi-carrier ejection techniques that are potentially able to overcome Shockley—Queisser Limit and Emerging PV

(Courtesy of Martin Green)

Type of PV systems

- PV arrays on spacecrafts and satellites
- Remote stand-alone without battery storage
- Remote stand-alone with battery storage
- Small modules for calculators and toys
- Residential grid connected with DC to AC inverter
- Commercial grid connected with inverters
- PV field for utility power generation

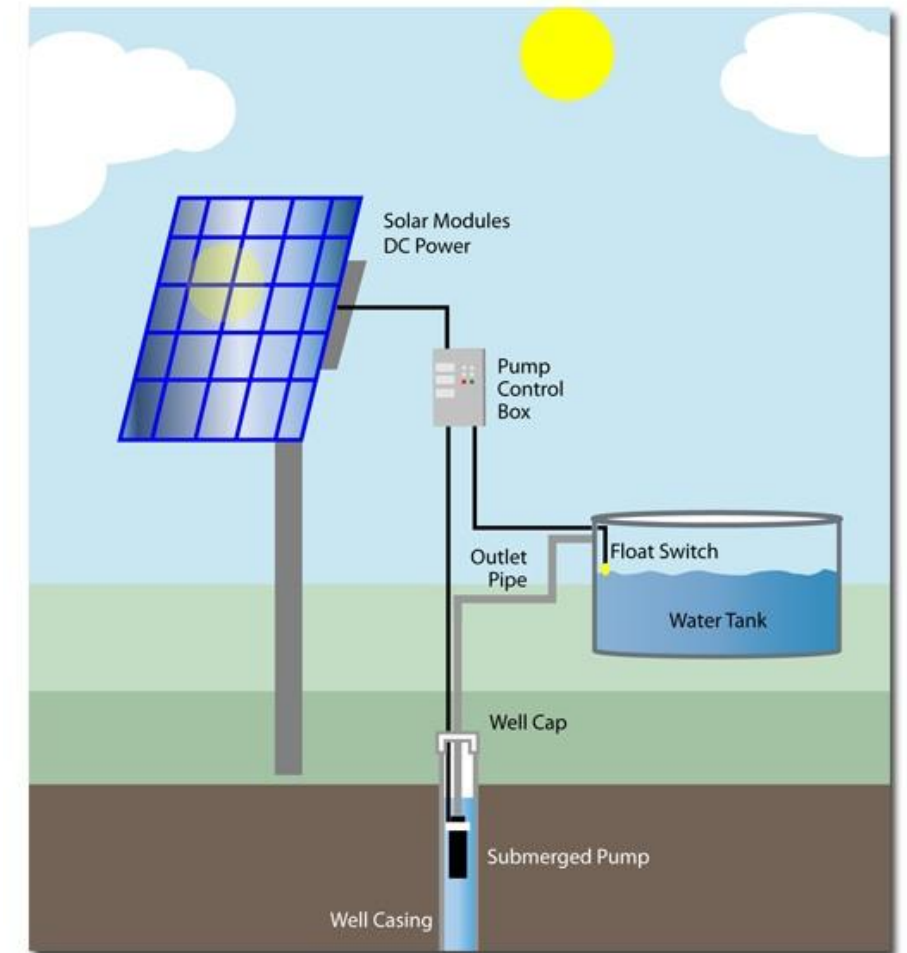
PV arrays on spacecrafts and satellites

- PV is a main power source for spacecraft operating in the inner Solar system.
- However, in the outer solar system, the sunlight is too weak to produce sufficient power and hence, radioisotope thermoelectric generators (RTGs) are used as power source.
- Multijunction solar cells are used for spacecrafts and satellites application because of high efficiency
- In 1971, the domestic solar cells were first used successfully on the Chinese satellite DFH-II.



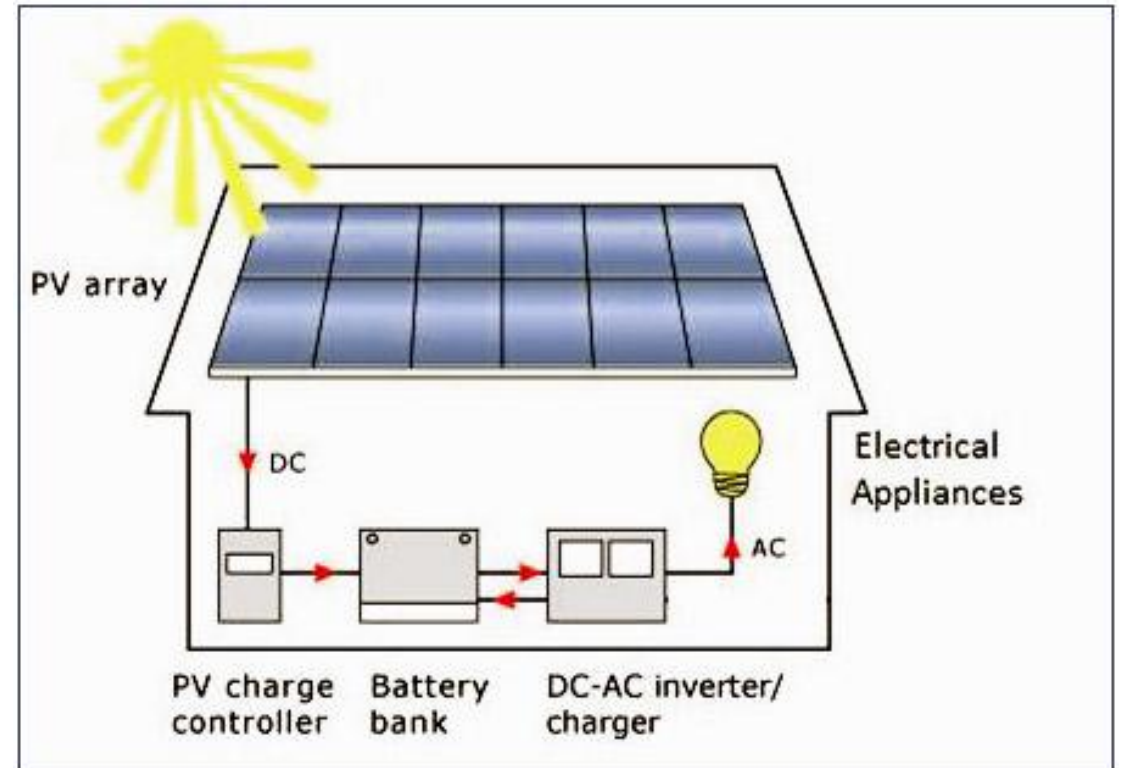
Remote stand-alone without battery storage

- Remote solar water pumping for crop irrigation, livestock watering and clean water for remote villages, especially for third world countries
- The underground water is pumped when the sun is shining
- Battery storage is not necessary
- Tracking system to maximize the usage time
- Simple system and economical
- Stand-alone system is also useful for air-conditioning in developed countries



Remote stand-alone with battery storage

- Use in remote applications where storage is needed to store electricity for when it is needed
- E.g. off-grid cabins, camps, site-specific energy for home
- Alternative systems: solar-wind combination, hybrid system (solar + AC generator)



Small modules for calculators, toys and wearable electronics

- The amorphous silicon is suitable for this application because only little current is required to run calculators.
- Flexible PV such as organic solar cell is suitable for wearable application.

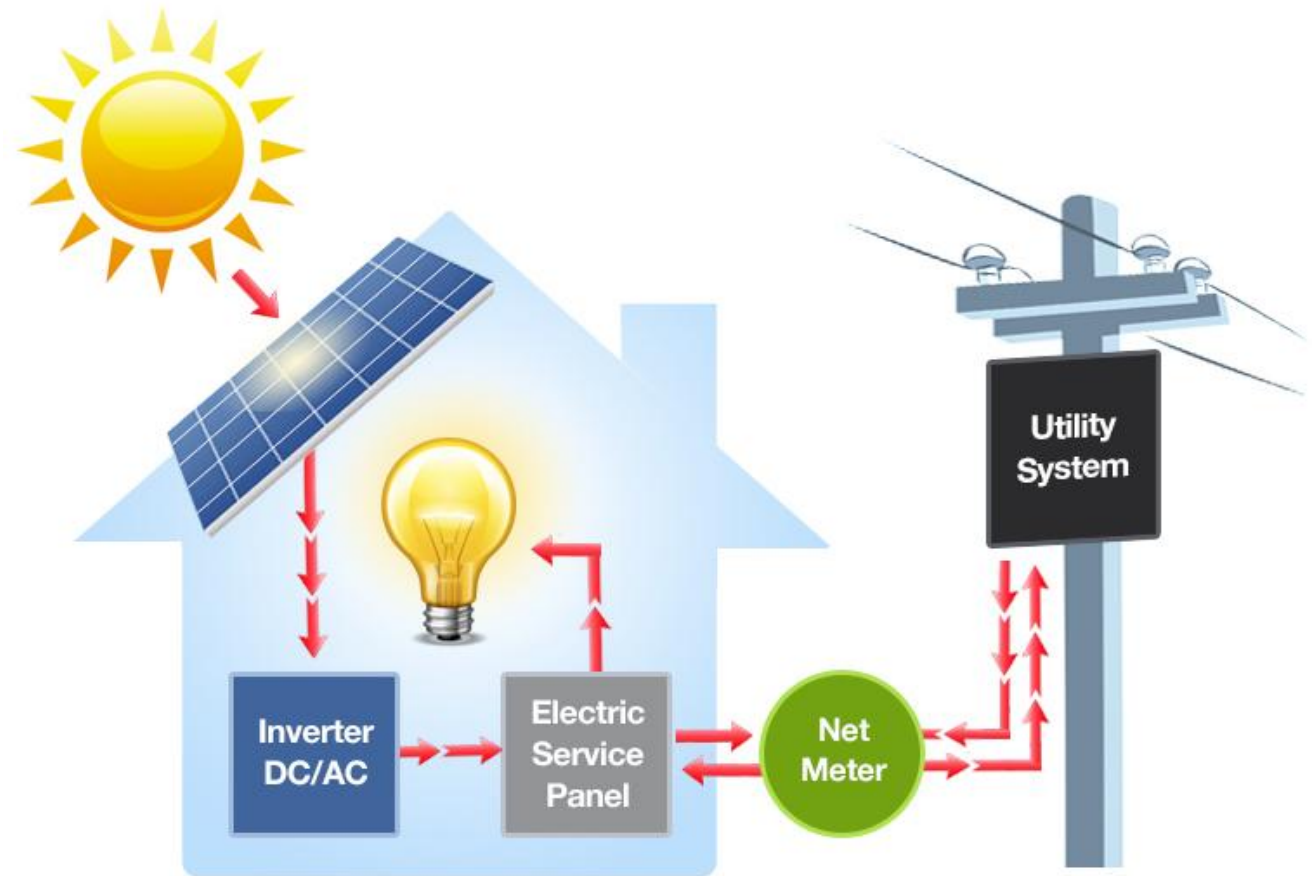


Tommy Hilfiger and Pvilion's
solar powered jacket



Residential grid connected with inverters

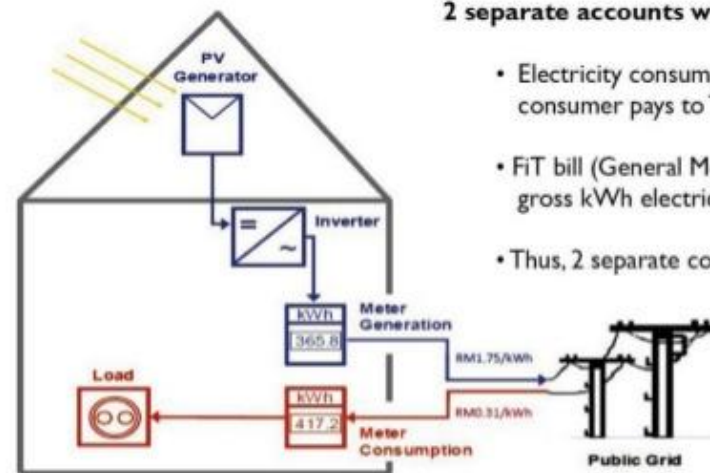
- The public policies such as **net metering** and **feed-in tariff** allows solar panel installers to sell the power to the utility at a favourable price.
- The net metering use a single, bi-directional meter that can measure current flow in two directions.
- <https://es-static-prod.s3.amazonaws.com/static/img/solar/solar-101/Net-metering-animation.388bae5d381a.gif>



Residential grid connected with inverters

- Feed-in tariff meter requires two separate meters for generation (sell) and consumption (purchase)

Feed In Tariff Concept



2 separate accounts with TNB:

- Electricity consumption bill (Consumption Meter): consumer pays to TNB for kWh electricity consumed
- FiT bill (General Meter): TNB pays to consumer for gross kWh electricity generated
- Thus, 2 separate contracts with TNB.

COMMERCIAL PRESENTATION

PV field (solar farm) for utility power generation

- While homeowners with solar panels on their roofs aim to generate enough power to cover their individual energy needs, large utility-scale solar farms are designed to generate enough electricity to power thousands of homes and businesses.
- With current PV technology, solar farming developers need approximately 2.5 acres of solar panels in high intensity sun areas to produce a megawatt of electricity; in moderate sun locations, about five acres of panels are needed to produce a megawatt.
- A megawatt is enough energy to power 650 residential homes for one day.



PV field (solar farm) for utility power generation



Longyangxia Dam solar farm (850 MW) was the world's largest solar farm from 2015 to 2017.

宁夏在腾格里沙漠南缘建成全国最大沙漠光伏集成区

作者：中国储能网新闻中心 来源：央广网 发布时间：2016-4-5 9:02:55

中国储能网讯：记者近日来到腾格里沙漠南缘的宁夏中卫市采访，了解到这里正在崛起全国最大的沙漠光伏电站，并把发展光伏和沙漠治理、节水农业相结合，开创了全国乃至全世界沙漠光伏并网电站的成功先河，对当地治理沙漠环境和发展新能源产业都具有十分重要的意义，兼顾了经济、环境和社会多种效益。



宁夏中卫过去因发明了草方格防风治沙技术享誉世界，现在当地沙漠上又建起了规模庞大的光伏电站集群，在茫茫大漠上构筑起“太阳能源长城”。记者在腾格里沙漠南缘看到成片成片的太阳能板延展开，一眼望去，宛如在金色的沙漠上悬起一抹蓝色的海洋。

中卫市腾格里沙漠东南缘有沙漠面积一千二百平方公里，这里的广阔沙漠就成为光伏发电的最好场所。为了充分利用和发挥中卫沙漠光热资源优势，大力发展新能源产业，中卫市于2012年开工建设沙漠光伏产业园，据介绍产业园规划面积六万亩，分为光伏发电区、光伏农业区、光伏制造产业区和观光旅游区四个部分，已建成20多座总规模1000多兆瓦的光伏电站，该光伏电站集成区每年可以新增上网20多亿度电，已形成从光伏发电、风能发电和光伏材料组件制造的全产业链。

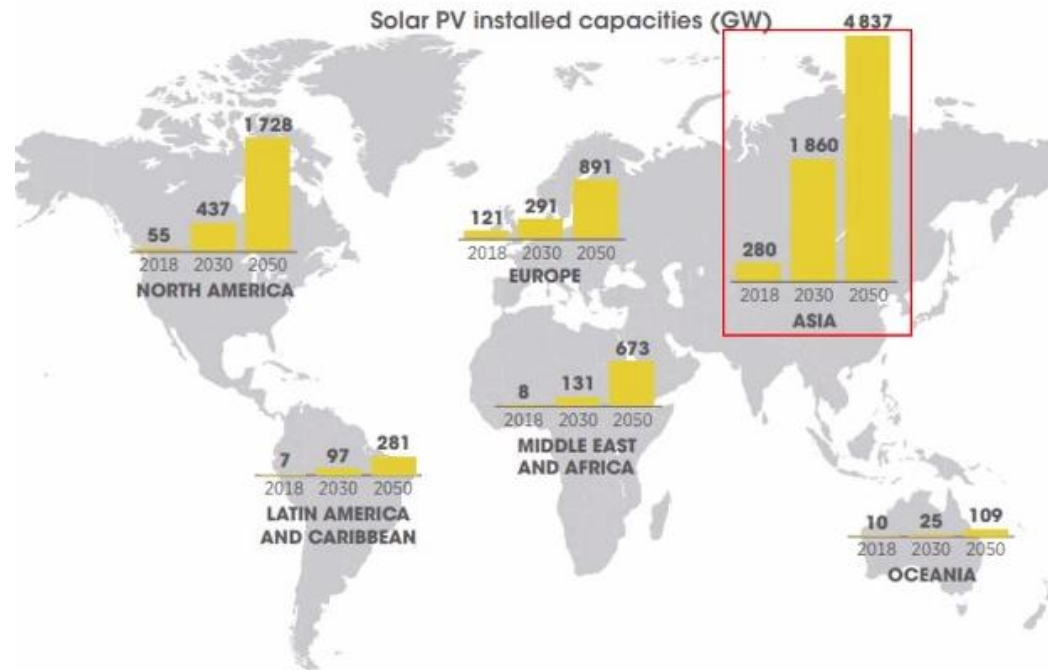
Tengger Desert solar farm, 1500MW capacity

World's largest photovoltaic power stations [\[edit \]](#)

Solar farms that are 150 MW or larger

Name	Country	Location	Capacity MW _p	Generation GWh p.a.	Size km ²	Year	Remarks	Ref
Tengger Desert Solar Park	 China	 37°33′00″N 105°03′14″E	1,547		43	2016	1,547 MW solar power was installed in Zhongwei, Ningxia by 2015.	[8]
Datong Solar Power Top Runner Base	 China	 40°04′25″N 113°08′12″E ,  40°00′19″N 112°57′20″E	1,000			2016	1 GW Phase I completed in June 2016. Total capacity will be 3 GW in 3 phases.	[9] [10] [11]
Kurnool Ultra Mega Solar Park	 India	 15.681522°N 78.283749°E	1,000		24	2017	Planned capacity 1000 MW	[12]
Longyangxia Dam Solar Park	 China	 36°10′54″N 100°34′41″E	850	824 ^{[13]}	23	2015	320 MW Phase I Completed in December 2013, 530 MW phase II in 2015	[14] [15] [16] [17] [18]
Kamuthi Solar Power Project	 India	 9°21′16″N 78°23′4″E	648		10.1	2016	Completed on 21 September 2016	[19] [20]
Solar Star (I and II)	 United States	 34°49′50″N 118°23′53″W	579		13	2015	579 MW _{AC} (747.3 MW _p) connected to the grid on June 19, 2015. ^{[21]} Consists of Solar Star I (318 MW _{AC} or 397.8 MW _{DC}) and Solar Star II: 279 MW _{AC} or 349.5 MW _{DC}	[22] [23]
Topaz Solar Farm	 United States	 35°23′N 120°4′W	550	1,301	24.6 ^{[24]}	2014	Gradually commissioned since February 2013, reached final capacity 550 MW in November 2014	[25] [26] [27]
Copper Mountain Solar Facility	 United States	 35°47′N 114°59′W	552			2015	Phase 1 completed in December 2010. Phase 2 completed in January 2013. Phase 3 completed in early 2015. Construction of 94 MW phase 4 completed in 2016.	[14] [28] [29] [30]
Desert Sunlight Solar Farm	 United States	 33°49′33″N 115°24′08″W	550	1,287	16	2015	Phase I with 300 MW completed in 2013. Construction of phase II to final capacity of 550 MW completed in January 2015	[31] [32] [33] [34] [35]
Huanghe Hydropower Golmud Solar Park	 China	 36°24′00″N 95°07′30″E	500		23	2014	Phase I completed in October 2011, followed by Phase II and III. 60 MW phase IV under construction. Within a group of 1,000 MW of co-located plants	[17] [36] [37] [38]

PV capacities by 2050



Chinese PV Industry Brief: China's NEA predicts 108 GW of solar in 2022

The Chinese government says the nation is on track to install 108 GW of PV this year. Huaneng says it will build a new 10 GW module factory, while Akcome has revealed plans to increase its heterojunction panel capacity by 6 GW.

MAY 31, 2022 VINCENT SHAW AND MAX HALL

COMMERCIAL & INDUSTRIAL PV MARKETS MODULES & UPSTREAM MANUFACTURING RESIDENTIAL PV UTILITY SCALE PV
UTILITY SCALE STORAGE CHINA

China dominated global solar jobs market in 2021

The 2.7 million people employed in solar in China last year dwarfed the 250,000 working in the US solar industry, according to a new report by the International Renewable Energy Agency.

SEPTEMBER 27, 2022 MAX HALL

EMPLOYMENT HIGHLIGHTS MARKETS MARKETS & POLICY WORLD



Image: IKEA

PV capacities will increase **20 times** around the world, especially in Asia, to promote carbon neutrality.

Take home message

- PV system
- History of solar cell
- Motivation
- Market history and policy
- Types of solar cells
- PV types and applications